NorthStar Distribution: Demand, Inventory and Transportation Analysis

Finding where the business is losing cash and capacity

Prepared by: Blessing Obot | Data Analyst

Profitability depends on selling the right products and moving them efficiently

- ✓FMCG distributor serving 100 retailers across three US regions.
- ✓Five product categories: Yogurt, Milk, SnackBar, ReadyMeal and Juice.
- ✓Objective: identify operational inefficiencies, quantify their impact and recommend improvements.

Three years of operational data reveal opportunities for improvement

- ✓ **Period:** January 2022 – December 2024
- ✓ **Data:** 190,754 sales records, 41,580 inventory records and 41,959 truck loads.
- ✓ **Tools:** SQL Server for data preparation and analysis; Power BI for visualisation.
- ✓ **Process:** Validate → Clean → Analyse → Visualise.
- ✓ Three invalid sales records were set aside.

Stalled sales make operational efficiency more important

–0.9% Units sold in 2024	$1.01M Excess inventory	39% Unused truck capacity

- ✓Revenue fell 1.0% in 2024.
- ✓Excess stock ties up cash while some products face stockout risk.
- ✓Underused truck capacity creates avoidable transportation costs.

Revenue has levelled off while truck utilisation remains low

- ✓ $19.95M revenue across three years.
- ✓ Quarterly revenue settled between $1.7M and $2.4M from 2023.
- ✓ Truck utilisation remained near 61% throughout the period.
- ✓ Yogurt generated $8.2M, approximately 41% of revenue.

BUSINESS PERFORMANCE OVERVIEW

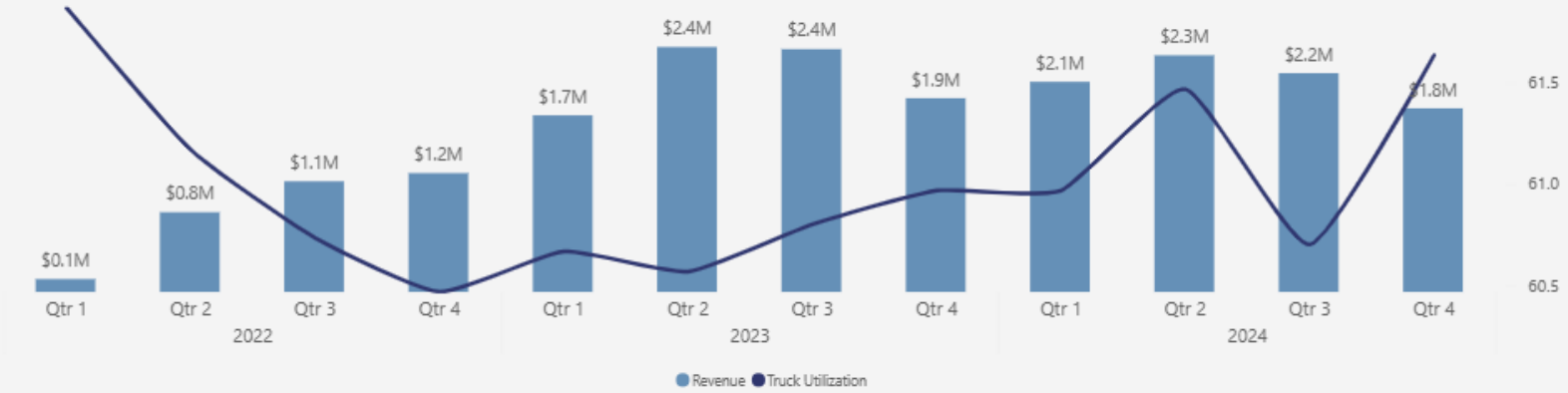
2022 2023 2024 Midwest Northeast South

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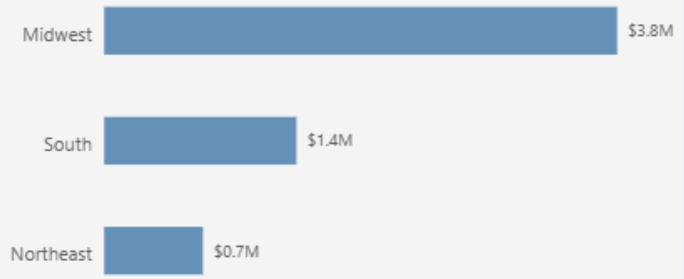


$19.95M	3.8M	389.81K	24.49M	$13.55M
TOTAL REVENUE	TOTAL UNIT SOLD	FORECAST Q1 2025	FUEL PURCHASED (Gallons)	ESTIMATED SAVINGS

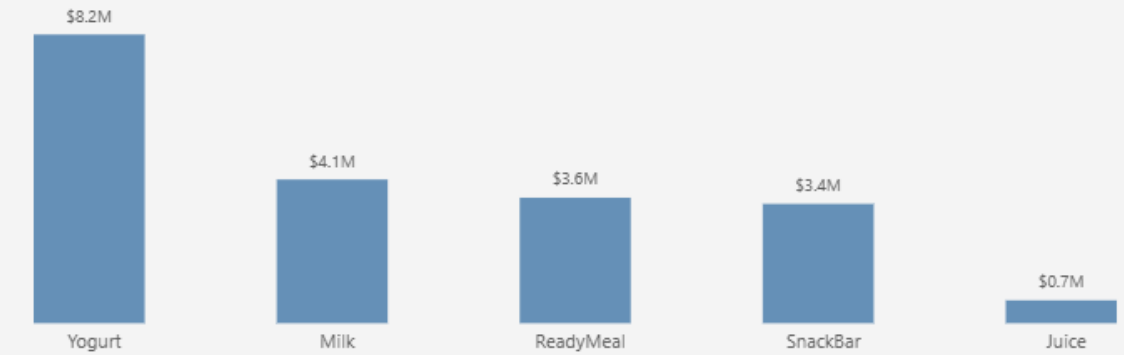
Revenue and Truck Utilization by Year



Fuel Cost on Empty Spaces by Region



Revenue by Category



Summary

NorthStar's growth has stalled, and three leaks are holding back performance: softening demand, cash tied up in the wrong stock, and trucks running about 39% empty.

- **Demand:** Units fell 0.9% and revenue 1.1% in 2024, with sales below 2023 from May to October. Q1 2025 is forecast at 389,810 units, 0.9% below Q1 2024.
- **Inventory:** $1.01M (20%) is excess, mainly Yogurt and SnackBar, while Milk and ReadyMeal stock out. About $994K can be released safely.
- **Transport:** Trucks stay near 61% full, and 48% of loads leave under 60% full, costing about $5.88M in fuel on empty space ($3.8M in the Midwest). Combining part-loaded trucks could save about $4.5M a year ($13.55M over 2022-2024), subject to a pilot. Separately, 40.2% of deliveries take four days or more, with no clear difference by region, channel or category.

Recommendation:
NorthStar should stop paying for stock and truck space it doesn't use. Pilot load consolidation in the Midwest, rebalance Yogurt and SnackBar stock to release cash, and fix Milk and ReadyMeal availability.

Sales fell below 2023 levels from May onwards

- ✓ January–April 2024 sales were 14.2% higher year-on-year.
- ✓ May–December sales were 7.2% lower.
- ✓ Full-year units fell 0.9%; revenue fell 1.0%.
- ✓ Q1 2025 demand is forecast at 389,810 units, down 0.9% year-on-year.

Excess stock and stockout risk exist side by side

- ✓ $5.15M inventory value; $1.01M classified as excess.
- ✓ Inventory covers approximately 72 days of demand.
- ✓ 21.5% of stock positions are at risk, based on recent stockouts or stock below reorder point.
- ✓ Yogurt holds the most excess stock; Milk and ReadyMeal have higher stockout rates.

DEMAND & INVENTORY PERFORMANCE

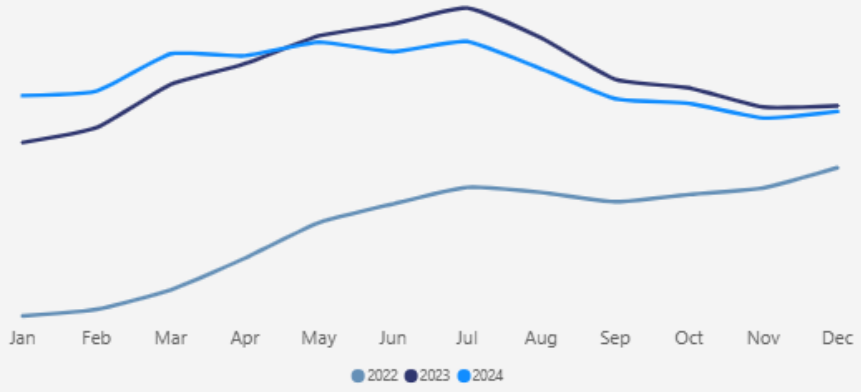
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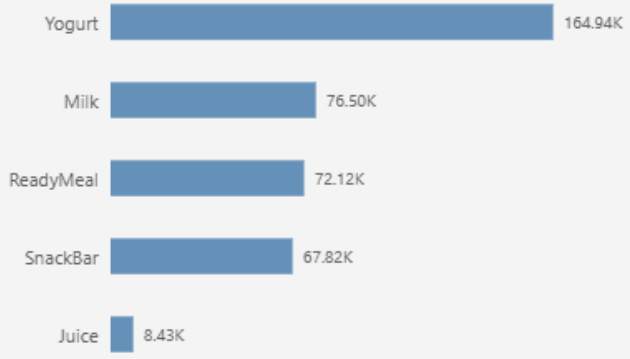


$5.15M	$1.01M	$993.97K	72.2	23.8%
INVENTORY VALUE	EXCESS INVENTORY	POTENTIAL CASH FREED	INVENTORY DAYS	STOCKOUT RATE

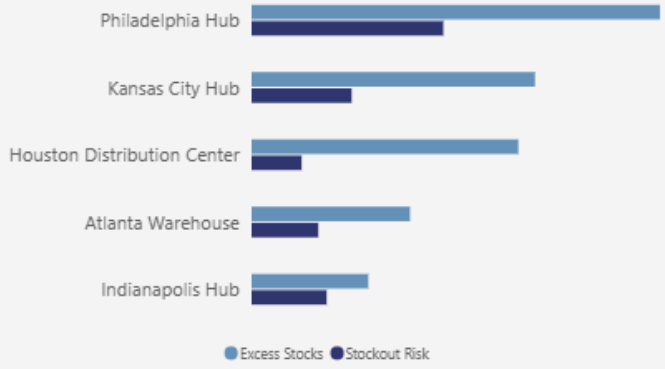
Monthly Units Sold by Year



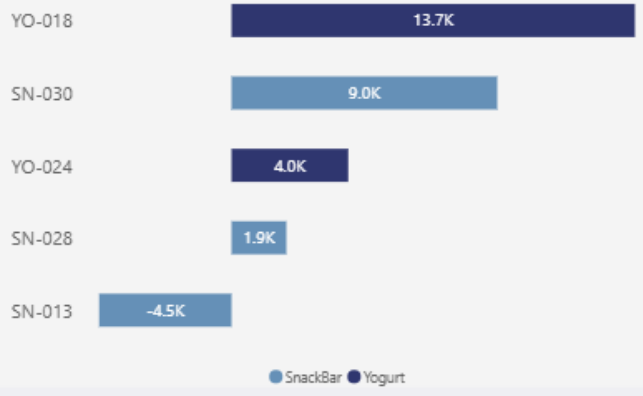
Q1 2025 Forecast by Category



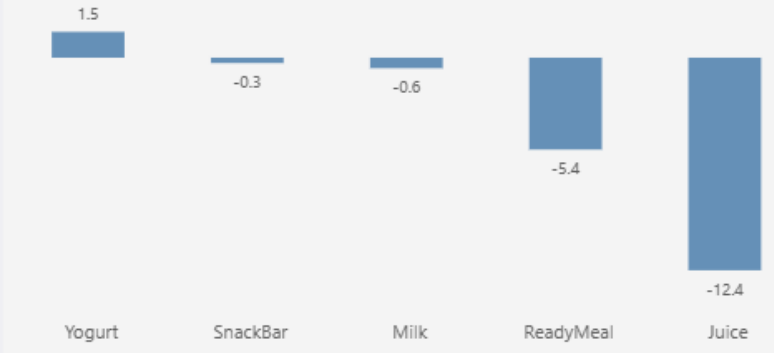
Stockout Risk and Excess by DC



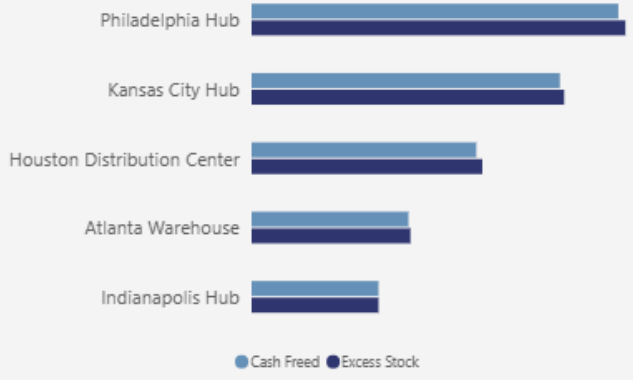
Change in Units by SKU



Forecast Growth by Category (%)



Excess Stock vs Cash Freed by DC



Nearly $994K in excess stock has no recent stockout history

- ✓ $994K in excess stock had no stockouts in the previous 52 weeks.
- ✓ Largest amounts: Philadelphia Hub $308K, Kansas City $259K, Houston $189K.
- ✓ Validate demand, supplier lead times and service requirements before reducing stock

Trucks average 61% utilisation, leaving 39% of capacity unused

- ✓ 48% of loads operate below 60% weight capacity.
- ✓ Midwest accounts for 64% of loads and approximately $3.8M in modelled fuel cost associated with unused capacity.
- ✓ Yogurt contributes the highest estimated unused-capacity fuel cost: $1.81M.

TRANSPORTATION OPERATIONS ANALYSIS

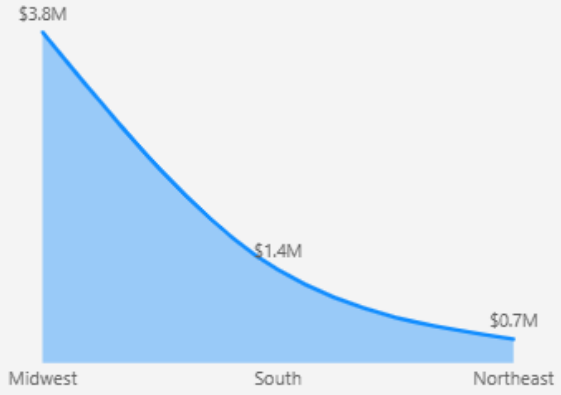
2022 2023 2024

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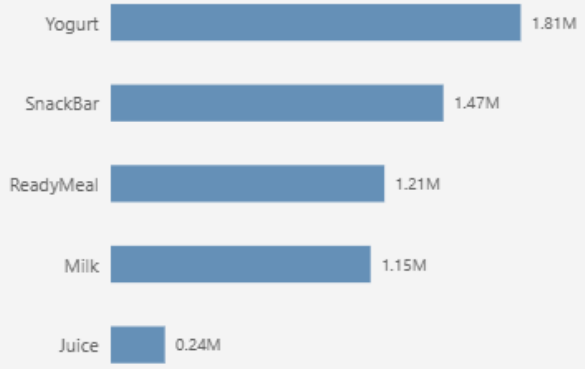


61.0%	$5.88M	3.0	40.2%	39.0%
TRUCK UTILIZATION	FUEL COST ON EMPTY SPACE	AVERAGE DELIVERY DAYS	DELAYED DELIVERIES	UNUSED TRUCK CAPACITY

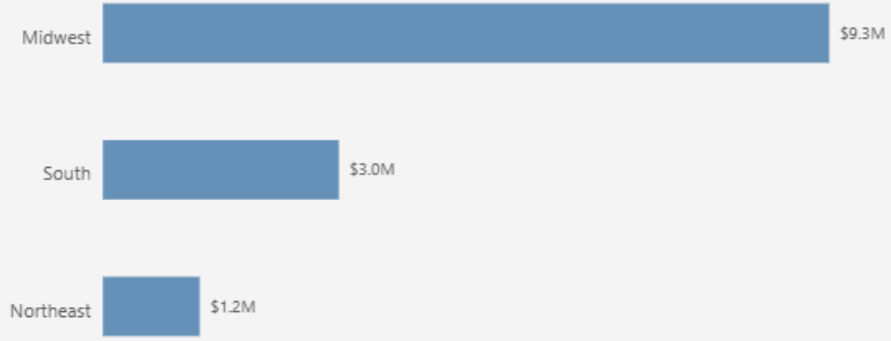
Fuel Cost by Region



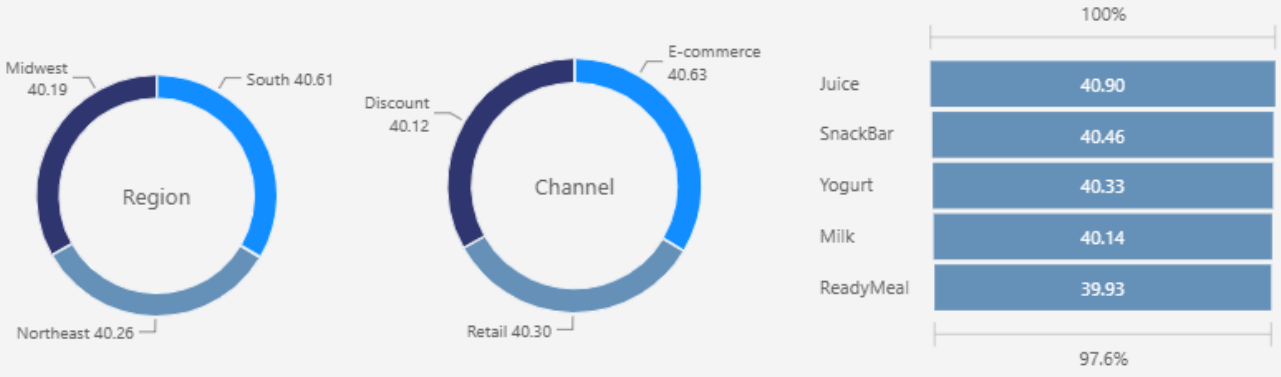
Wasted Fuel by Category



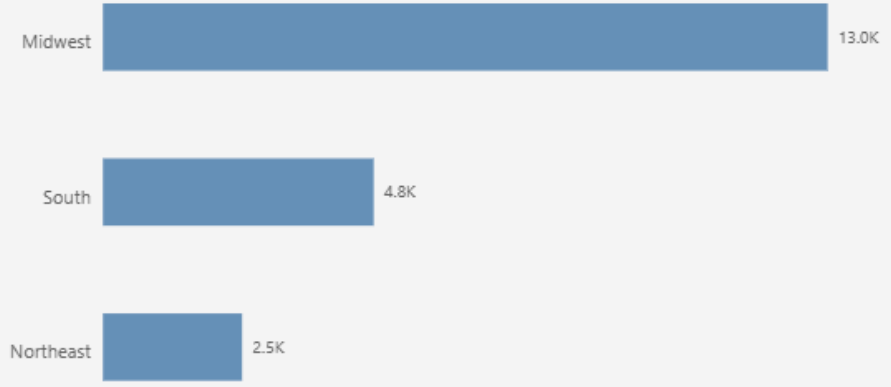
Estimated Savings by Region



Delayed Deliveries by Region, Channel & Category



Low-Utilization Loads by Region



Load consolidation could save an estimated $3.9M annually

- ✓ Modelled savings: $11.8M over three years, keeping refrigerated and ambient loads separate.
- ✓ Approximately $1M annually in fuel-only savings under the model.
- ✓ About 73% of the broader savings estimate depends on assumed non-fuel costs.
- ✓ Pilot compatible loads before scaling; actual savings will depend on operational constraints.

Four in ten delivery records show transit times of four days or more

- ✓ Average recorded delivery time: 3 days.
- ✓ 40.2% of records show delivery times of four days or longer.
- ✓ The pattern is evenly distributed across regions, channels and categories.
- ✓ Link order, dispatch and arrival timestamps to identify where delays occur.

Five actions can improve efficiency without relying on sales growth

1. Rebalance inventory to reduce excess stock and release working capital.
2. Review reorder levels for Milk and ReadyMeal to reduce stockouts.
3. Consolidate compatible loads, starting in the Midwest.
4. Pilot and measure actual cost, utilisation and delivery outcomes.
5. Improve planning using demand forecasts and monthly performance reviews.

Key estimates require operational validation

- ✓ Transportation costs use assumed mileage, fuel efficiency, payload and running costs.
- ✓ Consolidation estimates depend on route, timing and temperature compatibility.
- ✓ Delivery-time thresholds are not official service-level measures.
- ✓ Inventory costs are estimated at 70% of selling price; only 6 of 50 facilities are covered.
- ✓ Forecasts exclude promotions and external market changes.

Better planning can unlock cash and reduce avoidable costs

- ✓ Match inventory to actual demand.
- ✓ Improve truck utilisation through controlled consolidation.
- ✓ Investigate delivery delays using more detailed operational data.
- ✓ Measure results before expanding successful changes

THANK YOU

Questions or discussion